

Lute Lillo Portero

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Summary

I am a PhD student in Computer Science studying continual learning, loss of plasticity, and open-ended adaptation: why neural networks that perform well can gradually lose the ability to keep learning as tasks, data, and objectives shift. My research analyzes training dynamics, representations, and architectural mechanisms that preserve reusable learning structure, with the goal of building supervised and reinforcement-learning systems that remain adaptable over long, non-stationary learning horizons

Education

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| University of Vermont • Burlington, VT | 2024 – Present |
| • PhD in Computer Science | Advisor: Nick Cheney |
| • Graduate Research Assistant: continual learning, loss of plasticity, and open-ended learning. | |
| University of Nebraska – Omaha • Omaha, NE | Dec 2023 |
| • MSc in Computer Science. GPA: 4.0/4.0 | Advisor: Jorge Fandinno |
| • Research Assistant: logic programming / automated reasoning. | |
| University of Nebraska – Omaha • Omaha, NE | Dec 2022 |
| • BS in Computer Science. Minor in Mathematics. GPA: 3.9/4.0 | |

Work Experience

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| Software Engineer Intern , Amazon – Denver, CO | May 2023 – Aug 2023 |
| • Built back-end infrastructure enabling product variations in a bundle-widget feature (Java + JSP), supporting a large catalog scale (1M+ products). | |
| • Implemented front-end components for the feature using JavaScript and CSS. | |
| Software Engineer in Test Intern , Hudl – Lincoln, NE | May 2022 – Aug 2022 |
| • Designed and tested automated systems using Python and Selenium; worked with MongoDB-backed workflows. | |
| • Drafted and updated UI and API test designs in an Agile team and diagnosed failures in Sumo Logic. | |

Publications

- Lillo, Lute & Cheney, Nick. "On the Stability of Growth in Structural Plasticity". Preprint. [LINK](#)
- Lillo, Lute & Cheney, Nick. "Beyond Single-Model Optimization: Preserving Plasticity in Continual Reinforcement Learning". Accepted for publication at the Fifth Conference on Lifelong Learning Agents - CoLLAs 2026 (Oral). [LINK](#)
- Lillo, Lute & Cheney, Nick. "Activation Function Design Sustains Plasticity in Continual Learning". Proceedings of the International Conference on Learning Representations (ICLR) 2026. [LINK](#)
- Lillo, Lute & Cheney, Nick. "Heterosis-Based Crossover: Leveraging Structural Differences to Control Diversity and Bloat in Genetic Programming". GECCO'25 Companion: Proceedings of the Genetic and Evolutionary Computation Conference. [LINK](#)
- Fandinno, Jorge & Lillo, Lute. "Solving Epistemic Logic Programs using Generate-and-Test with Propagation". Proceedings of the 39th Annual AAAI Conference on Artificial Intelligence (Oral). [LINK](#)
- Lillo, Lute & Ali, H. "A Graph-Theoretic Approach for Examining Team Communities in Sports Transfer Markets". Proceedings of the 57th Hawaii International Conference on System Sciences (2024). [LINK](#)

Teaching

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| Teaching Assistant CSCI3320 (Data Structures), University of Nebraska at Omaha | Aug 2023 – Dec 2023 |
| Teaching Assistant CSCI4450 (Intro to AI), University of Nebraska at Omaha | Aug 2022 – Dec 2022 |

Skills & Additional Experience

ML / Research: continual learning; loss of plasticity; reinforcement learning; evolutionary computation; open-ended learning.

Programming: Python, Java, C, C#, Bash; Git; Linux.

ML Stack: PyTorch, JAX, TensorFlow, CUDA.

Languages: English and Spanish (bilingual).

Division I Men's Soccer: student-athlete for 5 collegiate seasons.